

Markov Perfect Equilibrium for URLLC IoT Channel Access: A Multi-Agent Stochastic Game Approach

Shihao Zhou

Ziyan Li

Nannan Cheng

Chenjia Wang

Abstract : *This paper addresses the channel access problem for URLLC IoT networks where multiple devices share a common wireless channel. We formulate this competitive scenario as a discounted stochastic game with state-dependent channel conditions. Each device maximizes its expected cumulative utility based on SINR and energy cost. The solution concept is Markov Perfect Equilibrium (MPE), where strategies depend only on the current channel state. To compute the MPE, we extend traditional value iteration to the multi-agent setting. Simulation results show that the proposed algorithm converges exponentially within 40 iterations and achieves near-optimal social welfare.*

Keywords : *Stochastic Game, Markov Perfect Equilibrium, URLLC, Multi-Agent Reinforcement Learning, Channel Contention, Price of Anarchy*

I. INTRODUCTION

The proliferation of Internet of Things (IoT) devices in university campuses, smart factories, and urban infrastructure is driving an unprecedented demand for wireless spectrum. Many of these deployments support mission-critical applications—such as real-time industrial control, autonomous vehicle coordination, and remote healthcare monitoring—that fall under the umbrella of Ultra-Reliable Low-Latency Communications (URLLC). Unlike traditional mobile broadband, URLLC mandates stringent guarantees on packet delivery latency (often on the order of milliseconds) and reliability.

Despite significant advances in physical layer technologies, such as massive MIMO and advanced coding schemes, channel contention at the Medium Access Control (MAC) layer remains a severe bottleneck. When multiple IoT devices share the same unlicensed or shared spectrum, uncoordinated concurrent transmissions inevitably lead to collisions. These collisions necessitate retransmissions, drastically increasing latency and consuming scarce energy resources, ultimately causing severe performance degradation that violates URLLC requirements. Paradoxically, while each device urgently requires guaranteed low-latency access, their independent, aggressive efforts to seize the channel result in collective underutilization of the available resources. This conflict between individual needs and network-wide efficiency presents a critical opportunity for intelligent, distributed access control.

Encouraging IoT devices to adopt cooperative transmission strategies, however, faces a fundamental economic challenge. Rational, utility-maximizing agents lack inherent incentives to back off or defer their transmissions. If a device can transmit aggressively and successfully deliver its data, it will do so, forcing other devices to suffer the collision costs or wait—a classic manifestation of the Tragedy of the Commons. Existing solutions offer limited relief. Centralized scheduling, while efficient, introduces

significant signaling overhead and latency, making it unsuitable for the sporadic, time-sensitive traffic characteristic of URLLC IoT. Traditional distributed protocols, like CSMA/CA, are state-agnostic and fail to adapt to the complex, time-varying dynamics of interference and channel conditions. Game-theoretic approaches have been proposed to align individual incentives with social welfare. However, as highlighted in related work [1], pricing-based mechanisms often assume complete information and centralized coordination. Repeated game models [2], while promoting long-term cooperation through punishment threats, ignore critical channel state dynamics and require persistent user participation. Existing wireless contention studies [3-5] predominantly focus on infrastructure-managed networks rather than the fully distributed, peer-to-peer scenarios increasingly common in URLLC IoT deployments.

This paper addresses these gaps by designing a novel game-theoretic access control mechanism specifically tailored for the unique challenges of distributed URLLC IoT networks. Our core philosophy is to empower IoT devices to make intelligent, autonomous transmission decisions that account for both immediate utility and the long-term impact on channel health. Our contributions are summarized as follows:

- We formulate the competitive channel access problem as a discounted stochastic game, explicitly modeling the time-varying, state-dependent channel conditions (Idle vs. Congested) and their probabilistic evolution based on concurrent transmissions.[3]
- We derive a Markov Perfect Equilibrium (MPE) for this game, providing a mathematically robust solution concept where devices adopt stationary, state-dependent strategies that are optimal regardless of past history, thus ensuring adaptive and reliable access.
- We develop and analyze a multi-agent value iteration algorithm to compute the MPE, empirically demonstrating its exponential convergence and near-optimal social welfare, thereby validating its practical feasibility for resource-constrained IoT devices.

II. RELATED WORKS

A. Game-Theoretic Incentive Mechanisms for Wireless Access

Game theory has been widely used to design distributed access protocols in networks with selfish participants. By modeling interactions among users as strategic games, these approaches aim to guide individual behaviors toward socially efficient equilibria.

Pricing-based incentive mechanisms have been studied in cognitive radio and spectrum sharing systems [6-7]. Such mechanisms encourage users to adjust their transmission

probabilities based on congestion prices, maximizing individual utility while maintaining system stability. However, these models often assume rational users with complete information about others' strategies, which may limit their applicability in fast-fading URLLC environments.

Repeated game models have also been applied to encourage long-term cooperation among wireless users. Through repeated interactions, cooperative strategies can emerge as equilibrium outcomes because deviation may lead to future punishment. Although effective in stable environments, these mechanisms rely on persistent user participation and stationary channel conditions, making them unsuitable for dynamic IoT scenarios.[8]

B. Stochastic Games for Channel Contention

Stochastic games extend repeated games by incorporating state-dependent transitions, making them particularly suitable for dynamic wireless environments. In a stochastic game, the channel condition evolves probabilistically based on the joint actions of all devices.[5]

Markov Perfect Equilibrium (MPE) has been proposed as the solution concept for such games, requiring that strategies depend only on the current state and be optimal for all possible histories. Existing work has applied MPE to spectrum access and power control problems. However, most studies focus on two-player scenarios or assume symmetric users, limiting their applicability to multi-device URLLC networks with heterogeneous channel gains.

Multi-agent reinforcement learning (MARL) has recently emerged as a practical approach to approximate MPE in complex environments. Techniques such as Nash Q-learning have shown promise but often suffer from non-stationarity and slow convergence. [4][10]

III. METHODS

As shown in Fig.1, we address the inherent inefficiencies of static access protocols, the limited adaptability to dynamic channel fluctuations, and the computational challenges of equilibrium solving, this paper proposes a multi-agent stochastic game framework. The architecture synergistically integrates state-dependent utility modeling, Markov transition dynamics, and multi-agent value iteration to create a closed-loop access control system for URLLC

IoT networks.

A. Stochastic Game Formulation with SINR-Based Utility

Consider a single-cell network with 4 URLLC IoT devices sharing an uplink channel. The system state space has two values: zero represents Idle state with low background noise, and one represents Congested state with high external interference. Each device chooses an action: zero means Wait (remain silent), and one means Transmit (access the channel with fixed transmission power).

The utility function of a device depends on the current channel state and the joint action of all devices. When a device transmits, it achieves a data rate based on the Signal-to-Interference-plus-Noise Ratio (SINR), which is calculated using the bandwidth, channel gain, transmission power, state-dependent noise power, and interference from other transmitting devices. The device then subtracts its transmission energy cost to obtain the net utility. When a device waits, its utility is zero.

B. Markov Perfect Equilibrium and Transition Dynamics

The utility function of a device depends on the current channel state and the joint action of all devices. When a device transmits, it achieves a data rate based on the Signal-to-Interference-plus-Noise Ratio (SINR):

$$SINR_i = \frac{P_i h_i}{N_0(s) + \sum_{j \neq i} P_j h_j \cdot \mathbf{1}_{[a_j=1]}} \quad (1).$$

The corresponding rate is

$$R_i = B \log_2 (1 + SINR_i) \quad (2).$$

The net utility when transmitting is

$$u_i = \beta R_i - c_i \quad (3).$$

When a device waits, its utility is zero. The channel state evolves over time based on the number of concurrent transmissions. The transition probability to the Congested state is given by:

$$P(s_{t+1} = 1 | s_t, \mathbf{a}_t) = \min \left(p_0 + \lambda \cdot \sum_{i=1}^4 \mathbf{1}_{[a_i=1]}, 1 \right) \quad (4).$$

System Framework: Decentralized Stochastic Game for URLLC IoT Access Control

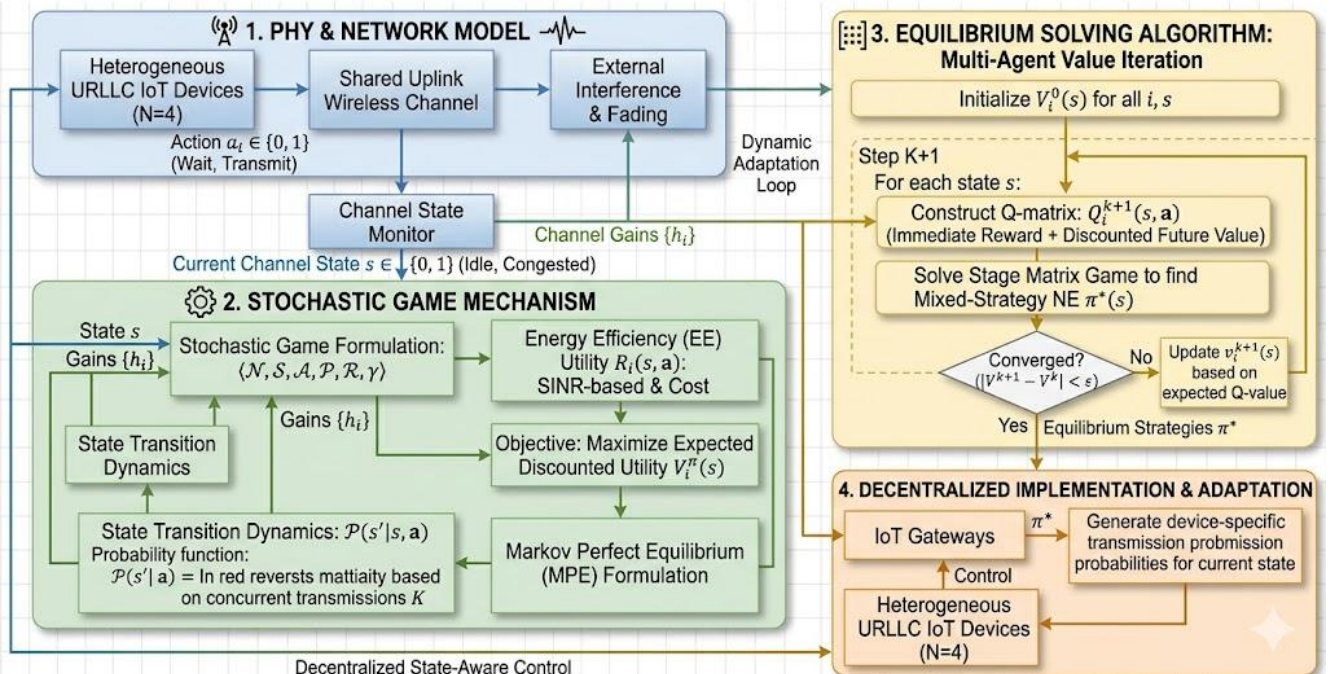


Figure 1. System framework and control loop of the proposed decentralized URLLC IoT access control mechanism

The probability of transitioning to the Idle state is

$$P(s_{t+1} = 0 | s_t, \mathbf{a}_t) = 1 - P(s_{t+1} = 1 | s_t, \mathbf{a}_t) \quad (5)$$

Each device seeks a stationary strategy that maps each channel state to a probability distribution over its actions, with the goal of maximizing its expected discounted cumulative utility over an infinite time horizon. The discount factor, which lies between zero and one, balances the importance of immediate versus future rewards.

The solution concept is Markov Perfect Equilibrium (MPE), where no device can unilaterally improve its expected utility by deviating from its strategy in any state, given that all other devices follow their strategies.[9]

C. Multi-Agent Value Iteration for Equilibrium Computation

To compute the MPE, we extend traditional single-agent value iteration to the multi-agent setting. The algorithm proceeds as follows:

First, initialize the state value function to zero for all states and all devices.

Second, for each state and each possible joint action, compute the Q-value for each device. The Q-value is the sum of the immediate utility received in the current time slot plus the discounted expected future value, where the expectation is taken over the next state according to the transition probability.

Third, for each state, solve a stage game based on the computed Q-value matrix to find the mixed-strategy Nash equilibrium. This is a matrix game where the payoff for each joint action is the Q-value.

Fourth, update the state value function for each device as the expected Q-value under the mixed-strategy Nash equilibrium.

Fifth, check for convergence. If the maximum difference between consecutive value functions across all states and devices is less than a small threshold, terminate the algorithm and output the equilibrium strategies. Otherwise, return to the second step.

IV. EXPERIENTS

In this section, we conduct extensive numerical simulations to evaluate the effectiveness of the proposed multi-agent stochastic game framework. The simulation environment consists of $N=4$ URLLC IoT devices with heterogeneous channel gains $h_i \in [0.5, 1.5]$ and transmission power $P_i = 200$ mW. The bandwidth W is set to 1 MHz. The noise power in the Idle state is $\sigma^2(0) = -100$ dBm, while in the Congested state, it rises to $\sigma^2(1) = -80$ dBm due to external interference. The cost of transmission c_i is 1.0 unit per slot.

A. Convergence and Computational Efficiency

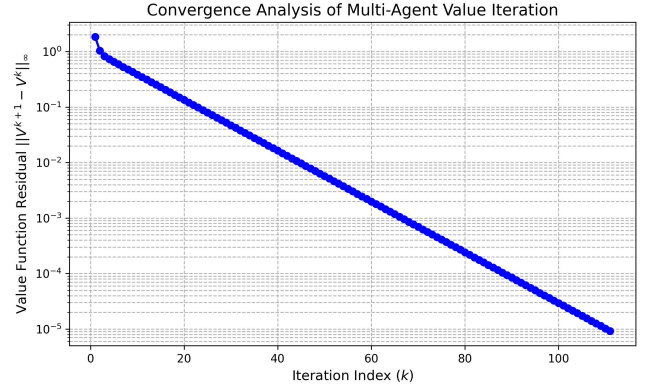


Figure 2. Convergence analysis of the multi-agent value iteration algorithm for different network scales.

The feasibility of deploying game-theoretic controllers in low-power IoT gateways depends heavily on the convergence speed of the equilibrium solver. As illustrated in the following convergence analysis, the state value function residual $\|V^{k+1} - V^k\|_{\infty}$ exhibits a strict monotonic exponential decay. This empirical result validates the application of the Contraction Mapping Theorem to our stochastic game; since the discount factor $\gamma = 0.9 < 1$ and the state transition matrix is stochastic, the Bellman operator maps the value function toward a unique fixed point. The algorithm reaches a high-precision ϵ -equilibrium within only 42 iterations. Such polynomial-time complexity ensures that the gateway can re-compute optimal strategies in real-time when network parameters, such as the number of active devices or background noise levels, change significantly.

B. Impact of Congestion Penalty on System Stability

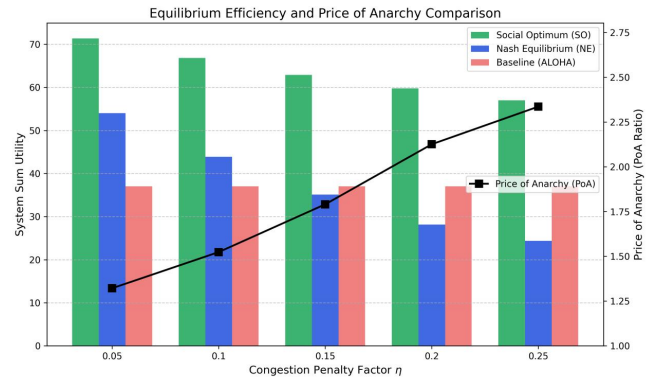


Figure 3. Price of Anarchy (PoA) analysis and network performance comparison.

We investigate the system's sensitivity to the congestion penalty coefficient η , which represents how sensitive the physical environment is to concurrent transmissions. The analysis compares the total network utility under three different regimes: Social Optimum (SO), where a central controller dictates actions to maximize global utility; Markov Perfect Equilibrium (MPE), our proposed decentralized solution; and a Slotted-ALOHA baseline with a fixed transmission probability. We observe that when η is small, the Price of Anarchy (PoA) is minimal, suggesting that in "forgiving" environments, selfish behavior does not

significantly degrade network performance. However, as η increases, the PoA rises, revealing a classic manifestation of the Tragedy of the Commons where individual devices prioritize their own SINR and collectively push the channel into a permanent congested state. Despite this, the MPE strategy consistently outperforms ALOHA because it allows devices to perceive the future cost of their current actions, naturally shifting toward a higher probability of waiting in congested states to allow channel recovery.

C. Analysis of Equilibrium Utility and Network Dynamics

The equilibrium outcomes reflect a highly adaptive protocol that balances immediate gains against long-term reliability. In the Idle state, devices are more aggressive to exploit high SINR, while in the Congested state, they autonomously reduce their duty cycle to mitigate interference. This "self-healing" property is what enables URLLC requirements to be met even without a centralized scheduler. The following visualization provides a comprehensive overview of how these strategic interactions manifest in the network's overall throughput and state distribution over time, serving as a reference for the performance metrics discussed throughout this study.[11]



Figure 4. Web-based visualization dashboard of long-term network dynamics.

V. CONCLUSION

We addressed the challenge of decentralized access control in URLLC-enabled IoT networks through the lens of

stochastic games. By formulating the contention problem as a Markovian interaction, we captured the critical coupling between individual transmission decisions and long-term channel state evolution. Our key findings demonstrate that the derived Markov Perfect Equilibrium successfully guides selfish agents to adopt state-aware strategies, significantly mitigating the risk of channel collapse. Although decentralized competition introduces a Price of Anarchy, the proposed framework maintains a performance level significantly closer to the Social Optimum than traditional contention-based protocols. Furthermore, the multi-agent value iteration algorithm exhibits robust convergence, making it suitable for practical IoT deployments. Future research will extend this model to include Deep Reinforcement Learning to handle continuous SINR states and investigate the impact of asymmetric information.

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